

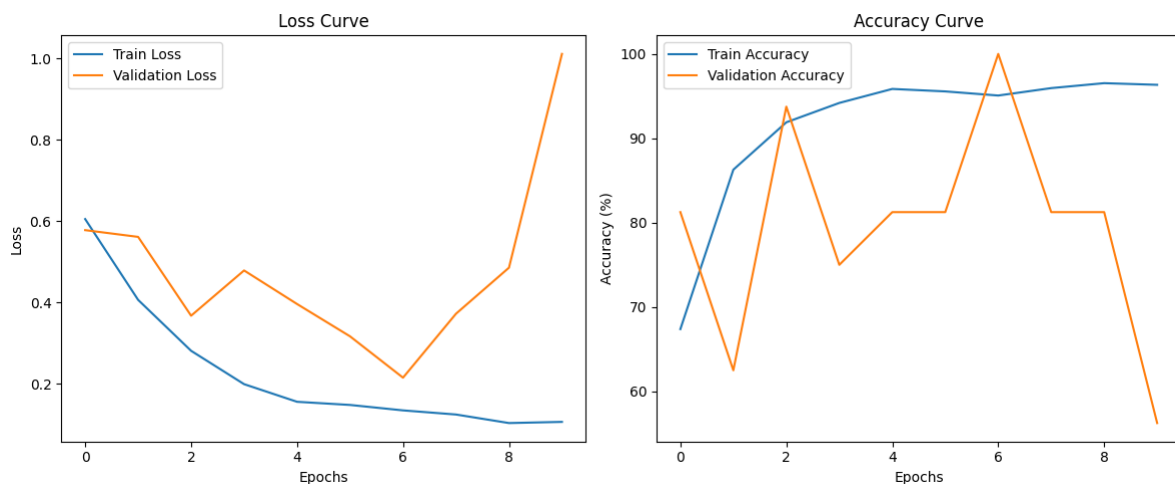
Model training, Data Augmentation, Hyper-parameter Tuning

I. 測試紀錄

Test 1 :

```
Sequential(  
  (0): Flatten(start_dim=1, end_dim=-1)  
  (1): Linear(in_features=65536, out_features=64, bias=True)  
  (2): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (3): ReLU()  
  (4): Dropout(p=0.5, inplace=False)  
  (5): Linear(in_features=64, out_features=64, bias=True)  
  (6): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (7): ReLU()  
  (8): Dropout(p=0.5, inplace=False)  
  (9): Linear(in_features=64, out_features=64, bias=True)  
  (10): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (11): ReLU()  
  (12): Dropout(p=0.5, inplace=False)  
  (13): Linear(in_features=64, out_features=1, bias=True)  
  (14): Sigmoid()  
)
```

```
Epoch 1/10 - loss: 0.6047 - train_acc: 67.38% - val_loss: 0.5774 - val_acc: 81.25% - time: 478.03s  
Epoch 2/10 - loss: 0.4059 - train_acc: 86.28% - val_loss: 0.5609 - val_acc: 62.50% - time: 60.94s  
Epoch 3/10 - loss: 0.2806 - train_acc: 91.89% - val_loss: 0.3670 - val_acc: 93.75% - time: 61.36s  
Epoch 4/10 - loss: 0.1986 - train_acc: 94.19% - val_loss: 0.4783 - val_acc: 75.00% - time: 60.74s  
Epoch 5/10 - loss: 0.1552 - train_acc: 95.85% - val_loss: 0.3957 - val_acc: 81.25% - time: 61.99s  
Epoch 6/10 - loss: 0.1475 - train_acc: 95.56% - val_loss: 0.3162 - val_acc: 81.25% - time: 60.43s  
Epoch 7/10 - loss: 0.1341 - train_acc: 95.07% - val_loss: 0.2146 - val_acc: 100.00% - time: 62.01s  
Epoch 8/10 - loss: 0.1240 - train_acc: 95.95% - val_loss: 0.3718 - val_acc: 81.25% - time: 60.59s  
Epoch 9/10 - loss: 0.1030 - train_acc: 96.53% - val_loss: 0.4851 - val_acc: 81.25% - time: 62.12s  
Epoch 10/10 - loss: 0.1058 - train_acc: 96.34% - val_loss: 1.0109 - val_acc: 56.25% - time: 60.69s
```



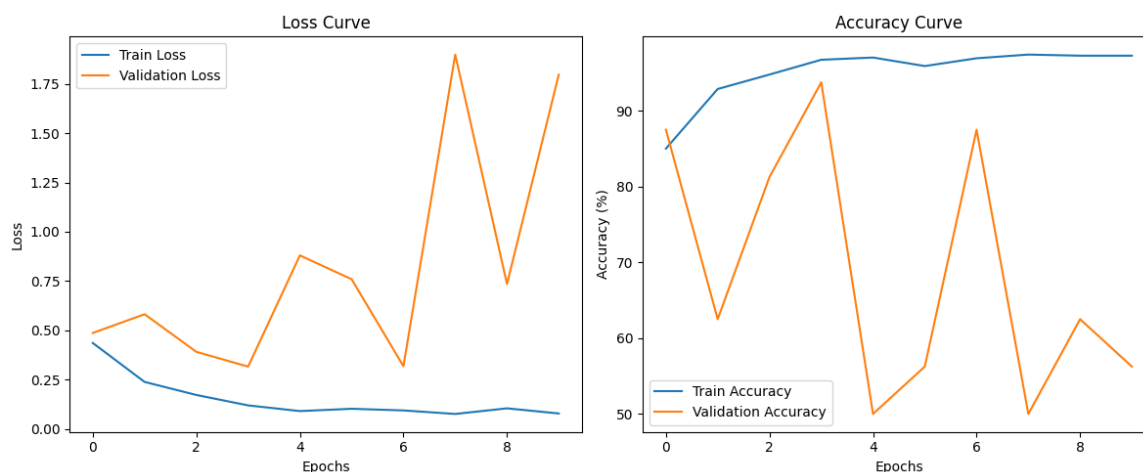
Test Accuracy: 78.65%
Test Loss: 0.4748

在全部參數維持預設值情況下，進行第一次執行，從摺線圖分析，train loss 穩定下降，train accuracy 穩定上升，但 validation loss 與 validation accuracy 後期表現過糟，波動情形偏大，需再改進。

Test 2 :

```
Sequential(  
  (0): Flatten(start_dim=1, end_dim=-1)  
  (1): Linear(in_features=65536, out_features=64, bias=True)  
  (2): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (3): ReLU()  
  (4): Dropout(p=0.3, inplace=False)  
  (5): Linear(in_features=64, out_features=64, bias=True)  
  (6): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (7): ReLU()  
  (8): Dropout(p=0.3, inplace=False)  
  (9): Linear(in_features=64, out_features=64, bias=True)  
  (10): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (11): ReLU()  
  (12): Dropout(p=0.3, inplace=False)  
  (13): Linear(in_features=64, out_features=1, bias=True)  
  (14): Sigmoid()  
)
```

```
Epoch 1/10 - loss: 0.4365 - train_acc: 85.01% - val_loss: 0.4873 - val_acc: 87.50% - time: 1539.08s  
Epoch 2/10 - loss: 0.2392 - train_acc: 92.87% - val_loss: 0.5822 - val_acc: 62.50% - time: 62.76s  
Epoch 3/10 - loss: 0.1731 - train_acc: 94.78% - val_loss: 0.3916 - val_acc: 81.25% - time: 62.16s  
Epoch 4/10 - loss: 0.1203 - train_acc: 96.73% - val_loss: 0.3167 - val_acc: 93.75% - time: 62.10s  
Epoch 5/10 - loss: 0.0915 - train_acc: 97.02% - val_loss: 0.8803 - val_acc: 50.00% - time: 63.37s  
Epoch 6/10 - loss: 0.1031 - train_acc: 95.90% - val_loss: 0.7595 - val_acc: 56.25% - time: 61.48s  
Epoch 7/10 - loss: 0.0948 - train_acc: 96.92% - val_loss: 0.3187 - val_acc: 87.50% - time: 62.66s  
Epoch 8/10 - loss: 0.0769 - train_acc: 97.41% - val_loss: 1.8979 - val_acc: 50.00% - time: 61.62s  
Epoch 9/10 - loss: 0.1054 - train_acc: 97.27% - val_loss: 0.7353 - val_acc: 62.50% - time: 62.87s  
Epoch 10/10 - loss: 0.0793 - train_acc: 97.27% - val_loss: 1.7960 - val_acc: 56.25% - time: 61.63s
```



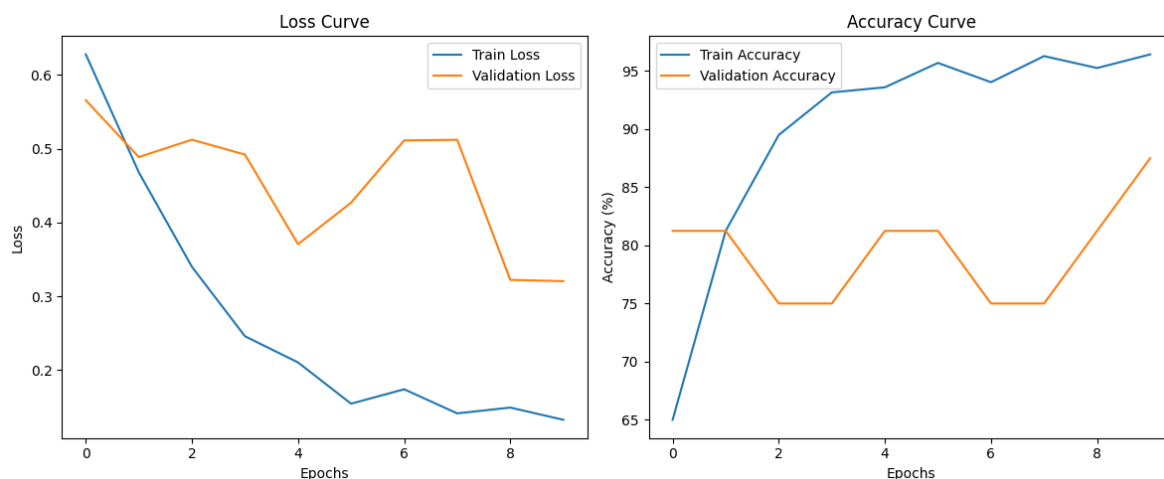
```
Test Accuracy: 75.05%  
Test Loss: 0.5577
```

調整 dropout 全為 0.3 後，train loss & accuracy 皆有更加良好表現，但 validation loss & accuracy 在後期皆有較劇烈波動，應為過擬合的表現，因此在 test 的結果也略微降低。

Test 3 :

```
Sequential(  
  (0): Flatten(start_dim=1, end_dim=-1)  
  (1): Linear(in_features=65536, out_features=64, bias=True)  
  (2): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (3): ReLU()  
  (4): Dropout(p=0.6, inplace=False)  
  (5): Linear(in_features=64, out_features=64, bias=True)  
  (6): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (7): ReLU()  
  (8): Dropout(p=0.6, inplace=False)  
  (9): Linear(in_features=64, out_features=64, bias=True)  
  (10): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (11): ReLU()  
  (12): Dropout(p=0.6, inplace=False)  
  (13): Linear(in_features=64, out_features=1, bias=True)  
  (14): Sigmoid()  
)
```

```
Epoch 1/10 - loss: 0.6278 - train_acc: 64.99% - val_loss: 0.5657 - val_acc: 81.25% - time: 980.66s  
Epoch 2/10 - loss: 0.4679 - train_acc: 81.25% - val_loss: 0.4887 - val_acc: 81.25% - time: 63.03s  
Epoch 3/10 - loss: 0.3401 - train_acc: 89.50% - val_loss: 0.5123 - val_acc: 75.00% - time: 61.96s  
Epoch 4/10 - loss: 0.2460 - train_acc: 93.16% - val_loss: 0.4920 - val_acc: 75.00% - time: 62.94s  
Epoch 5/10 - loss: 0.2105 - train_acc: 93.60% - val_loss: 0.3707 - val_acc: 81.25% - time: 61.60s  
Epoch 6/10 - loss: 0.1546 - train_acc: 95.70% - val_loss: 0.4270 - val_acc: 81.25% - time: 62.87s  
Epoch 7/10 - loss: 0.1742 - train_acc: 94.04% - val_loss: 0.5112 - val_acc: 75.00% - time: 61.78s  
Epoch 8/10 - loss: 0.1415 - train_acc: 96.29% - val_loss: 0.5121 - val_acc: 75.00% - time: 63.33s  
Epoch 9/10 - loss: 0.1494 - train_acc: 95.26% - val_loss: 0.3224 - val_acc: 81.25% - time: 61.63s  
Epoch 10/10 - loss: 0.1329 - train_acc: 96.44% - val_loss: 0.3206 - val_acc: 87.50% - time: 62.33s
```



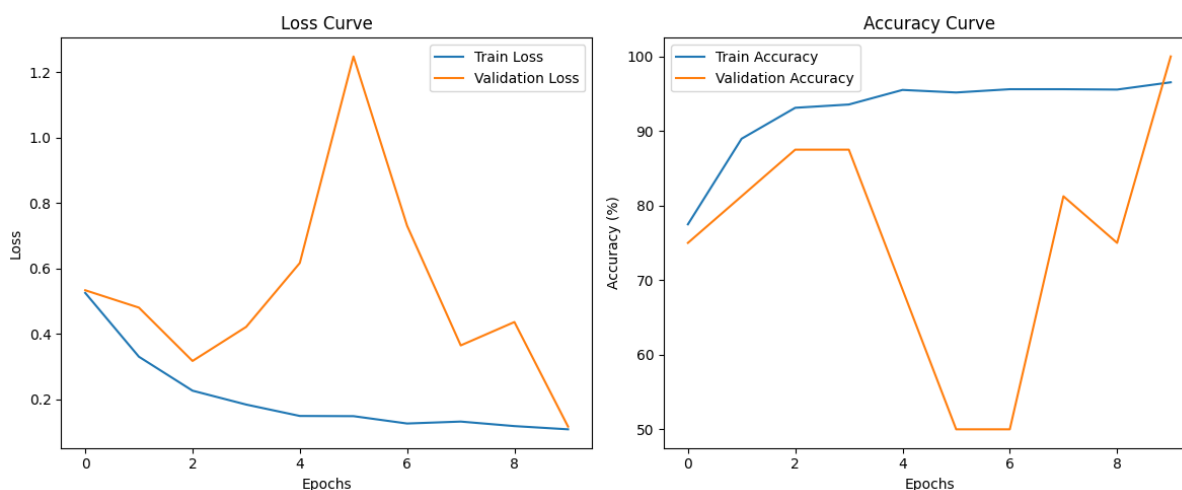
Test Accuracy: 82.08%
Test Loss: 0.4595

調整 dropout 全為 0.6 後，運算耗費時間大幅減少，train loss & accuracy 後期還是可以達到良好表現，又 validation loss & accuracy 在後期穩定性相較 Test 1(dropout 全為 0.5)好，因此最後得到的 test accuracy 略微提升。

Test 4 :

```
Sequential(  
  (0): Flatten(start_dim=1, end_dim=-1)  
  (1): Linear(in_features=65536, out_features=64, bias=True)  
  (2): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (3): ReLU()  
  (4): Dropout(p=0.6, inplace=False)  
  (5): Linear(in_features=64, out_features=64, bias=True)  
  (6): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (7): ReLU()  
  (8): Dropout(p=0.4, inplace=False)  
  (9): Linear(in_features=64, out_features=64, bias=True)  
  (10): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (11): ReLU()  
  (12): Dropout(p=0.3, inplace=False)  
  (13): Linear(in_features=64, out_features=1, bias=True)  
  (14): Sigmoid()  
)
```

```
Epoch 1/10 - loss: 0.5256 - train_acc: 77.49% - val_loss: 0.5337 - val_acc: 75.00% - time: 64.06s  
Epoch 2/10 - loss: 0.3306 - train_acc: 88.96% - val_loss: 0.4807 - val_acc: 81.25% - time: 61.91s  
Epoch 3/10 - loss: 0.2267 - train_acc: 93.12% - val_loss: 0.3176 - val_acc: 87.50% - time: 63.28s  
Epoch 4/10 - loss: 0.1843 - train_acc: 93.55% - val_loss: 0.4217 - val_acc: 87.50% - time: 61.91s  
Epoch 5/10 - loss: 0.1494 - train_acc: 95.51% - val_loss: 0.6171 - val_acc: 68.75% - time: 63.12s  
Epoch 6/10 - loss: 0.1488 - train_acc: 95.17% - val_loss: 1.2486 - val_acc: 50.00% - time: 61.78s  
Epoch 7/10 - loss: 0.1264 - train_acc: 95.61% - val_loss: 0.7309 - val_acc: 50.00% - time: 63.17s  
Epoch 8/10 - loss: 0.1323 - train_acc: 95.61% - val_loss: 0.3651 - val_acc: 81.25% - time: 61.64s  
Epoch 9/10 - loss: 0.1184 - train_acc: 95.56% - val_loss: 0.4366 - val_acc: 75.00% - time: 62.82s  
Epoch 10/10 - loss: 0.1086 - train_acc: 96.53% - val_loss: 0.1169 - val_acc: 100.00% - time: 61.46s
```



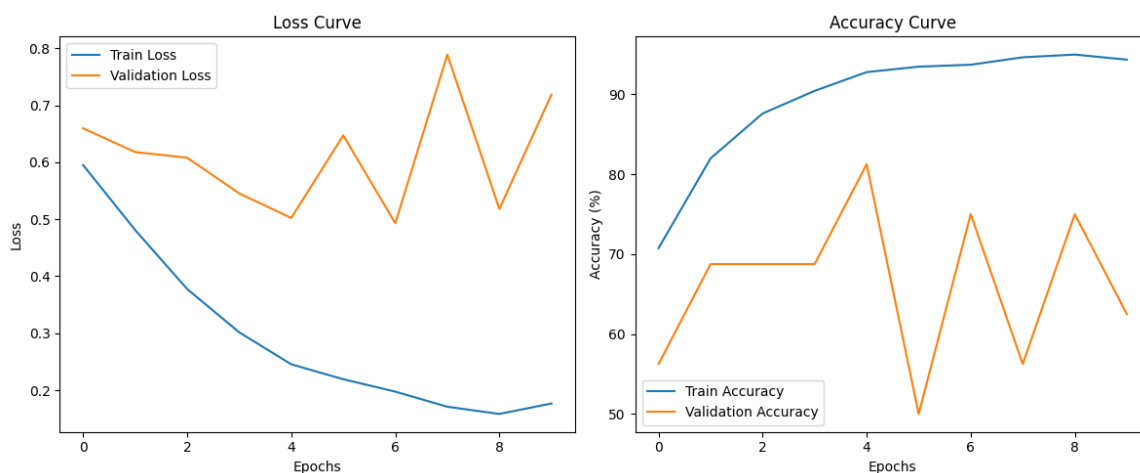
Test Accuracy: 83.18%
Test Loss: 0.4209

調整 dropout 依序分別為 0.6、0.4、0.3，train loss & accuracy 表現穩定，然而 validation loss & accuracy 在某些 epoch 波動過大，穩定性不佳，最後得到的 test accuracy 相較前一次 (Test 3)些微提升。

Test 5 :

```
Sequential(  
  (0): Flatten(start_dim=1, end_dim=-1)  
  (1): Linear(in_features=65536, out_features=32, bias=True)  
  (2): BatchNorm1d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (3): ReLU()  
  (4): Dropout(p=0.6, inplace=False)  
  (5): Linear(in_features=32, out_features=32, bias=True)  
  (6): BatchNorm1d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (7): ReLU()  
  (8): Dropout(p=0.4, inplace=False)  
  (9): Linear(in_features=32, out_features=32, bias=True)  
  (10): BatchNorm1d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (11): ReLU()  
  (12): Dropout(p=0.3, inplace=False)  
  (13): Linear(in_features=32, out_features=1, bias=True)  
  (14): Sigmoid()  
)
```

```
Epoch 1/10 - loss: 0.5950 - train_acc: 70.75% - val_loss: 0.6595 - val_acc: 56.25% - time: 63.71s  
Epoch 2/10 - loss: 0.4811 - train_acc: 81.98% - val_loss: 0.6179 - val_acc: 68.75% - time: 63.69s  
Epoch 3/10 - loss: 0.3775 - train_acc: 87.60% - val_loss: 0.6080 - val_acc: 68.75% - time: 61.73s  
Epoch 4/10 - loss: 0.3015 - train_acc: 90.43% - val_loss: 0.5454 - val_acc: 68.75% - time: 63.06s  
Epoch 5/10 - loss: 0.2455 - train_acc: 92.77% - val_loss: 0.5023 - val_acc: 81.25% - time: 61.54s  
Epoch 6/10 - loss: 0.2192 - train_acc: 93.46% - val_loss: 0.6472 - val_acc: 50.00% - time: 62.76s  
Epoch 7/10 - loss: 0.1975 - train_acc: 93.70% - val_loss: 0.4930 - val_acc: 75.00% - time: 61.77s  
Epoch 8/10 - loss: 0.1710 - train_acc: 94.63% - val_loss: 0.7890 - val_acc: 56.25% - time: 62.85s  
Epoch 9/10 - loss: 0.1583 - train_acc: 94.97% - val_loss: 0.5180 - val_acc: 75.00% - time: 61.99s  
Epoch 10/10 - loss: 0.1767 - train_acc: 94.34% - val_loss: 0.7185 - val_acc: 62.50% - time: 62.76s
```



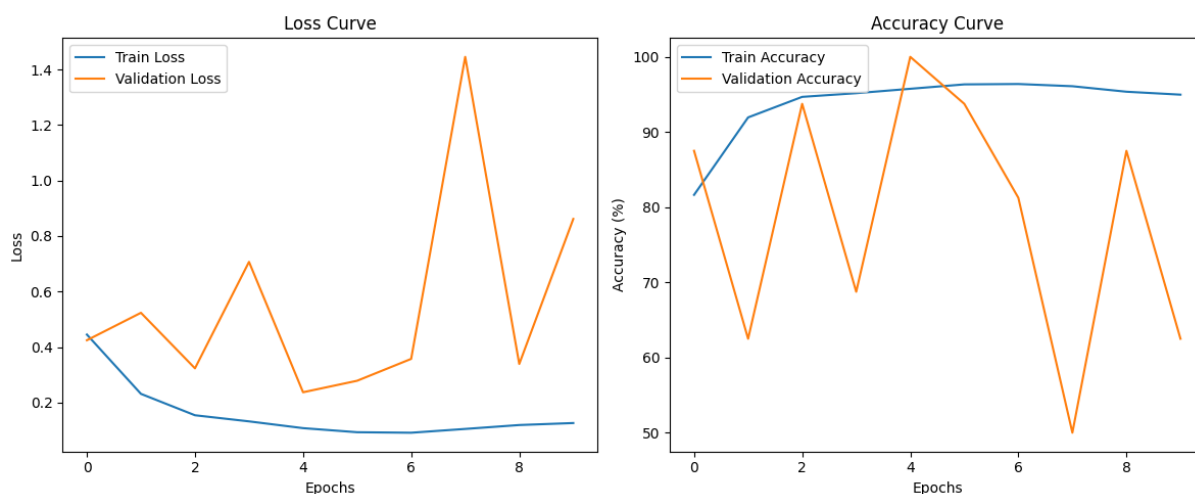
```
Test Accuracy: 76.77%  
Test Loss: 0.4495
```

維持 dropout 分別依序為 0.6、0.4、0.3，同時一併減少每層計算的神經元數，從 64 降為 32，train loss & accuracy 表現漸進趨於穩定，但 validation loss & accuracy 明顯偏差過大，應為欠擬合的情形，最後得到的 test accuracy 也不夠理想。

Test 6 :

```
Sequential(  
  (0): Flatten(start_dim=1, end_dim=-1)  
  (1): Linear(in_features=65536, out_features=128, bias=True)  
  (2): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (3): ReLU()  
  (4): Dropout(p=0.6, inplace=False)  
  (5): Linear(in_features=128, out_features=128, bias=True)  
  (6): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (7): ReLU()  
  (8): Dropout(p=0.4, inplace=False)  
  (9): Linear(in_features=128, out_features=128, bias=True)  
  (10): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (11): ReLU()  
  (12): Dropout(p=0.3, inplace=False)  
  (13): Linear(in_features=128, out_features=1, bias=True)  
  (14): Sigmoid()  
)
```

```
Epoch 1/10 - loss: 0.4449 - train_acc: 81.64% - val_loss: 0.4244 - val_acc: 87.50% - time: 63.14s  
Epoch 2/10 - loss: 0.2313 - train_acc: 91.94% - val_loss: 0.5233 - val_acc: 62.50% - time: 63.32s  
Epoch 3/10 - loss: 0.1541 - train_acc: 94.68% - val_loss: 0.3231 - val_acc: 93.75% - time: 63.21s  
Epoch 4/10 - loss: 0.1321 - train_acc: 95.17% - val_loss: 0.7067 - val_acc: 68.75% - time: 61.92s  
Epoch 5/10 - loss: 0.1077 - train_acc: 95.75% - val_loss: 0.2370 - val_acc: 100.00% - time: 63.08s  
Epoch 6/10 - loss: 0.0931 - train_acc: 96.34% - val_loss: 0.2784 - val_acc: 93.75% - time: 61.94s  
Epoch 7/10 - loss: 0.0912 - train_acc: 96.39% - val_loss: 0.3572 - val_acc: 81.25% - time: 62.92s  
Epoch 8/10 - loss: 0.1049 - train_acc: 96.09% - val_loss: 1.4453 - val_acc: 50.00% - time: 62.66s  
Epoch 9/10 - loss: 0.1190 - train_acc: 95.36% - val_loss: 0.3389 - val_acc: 87.50% - time: 63.00s  
Epoch 10/10 - loss: 0.1261 - train_acc: 94.97% - val_loss: 0.8615 - val_acc: 62.50% - time: 62.62s
```



```
Test Accuracy: 81.93%  
Test Loss: 0.4333
```

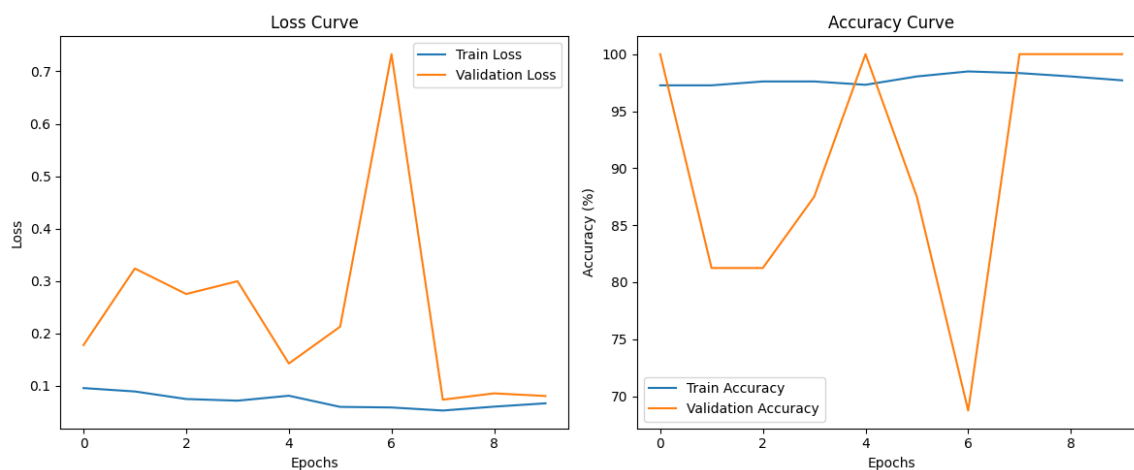
維持 dropout 分別依序為 0.6、0.4、0.3，改為一併增加每層計算的神經元數，從 64 增為 128，train loss & accuracy 表現穩定良好，但 validation loss & accuracy 波動情形過大，極為不穩定，應為過擬合的情形。

Test 7 :

```
Sequential(  
  (0): Flatten(start_dim=1, end_dim=-1)  
  (1): Linear(in_features=65536, out_features=128, bias=True)  
  (2): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (3): ReLU()  
  (4): Dropout(p=0.6, inplace=False)  
  (5): Linear(in_features=128, out_features=128, bias=True)  
  (6): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (7): ReLU()  
  (8): Dropout(p=0.4, inplace=False)  
  (9): Linear(in_features=128, out_features=128, bias=True)  
  (10): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (11): ReLU()  
  (12): Dropout(p=0.3, inplace=False)  
  (13): Linear(in_features=128, out_features=1, bias=True)  
  (14): Sigmoid()  
)
```

```
# hyperparameter  
lr = 0.0001  
weight_decay = 0.001  
epochs = 10
```

```
Epoch 1/10 - loss: 0.0956 - train_acc: 97.27% - val_loss: 0.1777 - val_acc: 100.00% - time: 63.70s  
Epoch 2/10 - loss: 0.0891 - train_acc: 97.27% - val_loss: 0.3238 - val_acc: 81.25% - time: 61.64s  
Epoch 3/10 - loss: 0.0747 - train_acc: 97.61% - val_loss: 0.2751 - val_acc: 81.25% - time: 63.62s  
Epoch 4/10 - loss: 0.0716 - train_acc: 97.61% - val_loss: 0.2995 - val_acc: 87.50% - time: 62.20s  
Epoch 5/10 - loss: 0.0810 - train_acc: 97.31% - val_loss: 0.1425 - val_acc: 100.00% - time: 63.86s  
Epoch 6/10 - loss: 0.0597 - train_acc: 98.05% - val_loss: 0.2128 - val_acc: 87.50% - time: 63.52s  
Epoch 7/10 - loss: 0.0586 - train_acc: 98.49% - val_loss: 0.7329 - val_acc: 68.75% - time: 62.33s  
Epoch 8/10 - loss: 0.0527 - train_acc: 98.34% - val_loss: 0.0735 - val_acc: 100.00% - time: 63.04s  
Epoch 9/10 - loss: 0.0602 - train_acc: 98.05% - val_loss: 0.0855 - val_acc: 100.00% - time: 61.44s  
Epoch 10/10 - loss: 0.0665 - train_acc: 97.71% - val_loss: 0.0805 - val_acc: 100.00% - time: 63.62s
```



```
Test Accuracy: 83.85%  
Test Loss: 0.4179
```

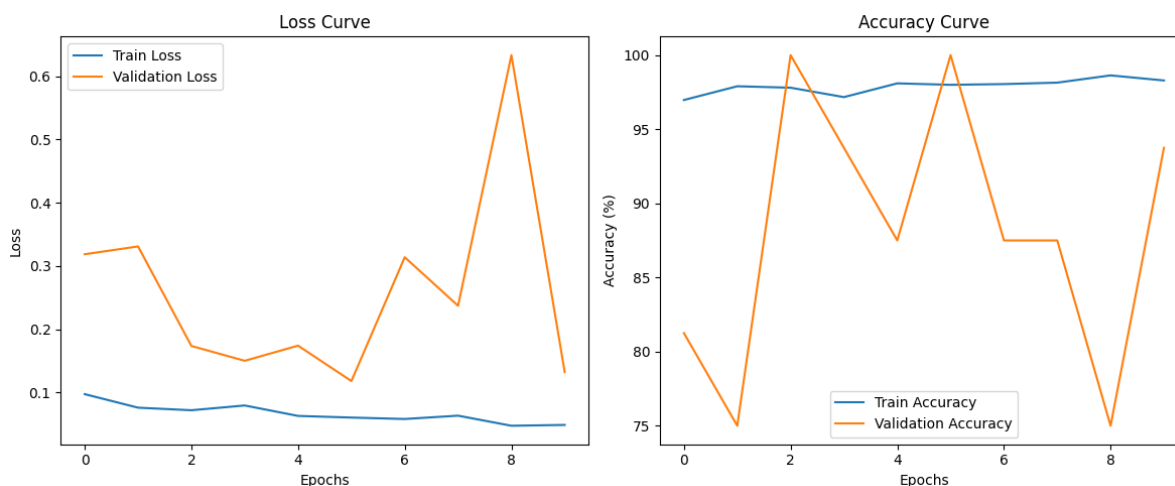
維持前一次(**Test 6**) model 設定，唯獨改變 learning rate 為 0.0001，train loss & accuracy 穩定度大幅提升，但 validation loss & accuracy 波動情形仍然過大，極為不穩定，應為過擬合的情形，需再改進，而得到結果 test accuracy 則為目前最佳。

Test 8 :

```
Sequential(  
  (0): Flatten(start_dim=1, end_dim=-1)  
  (1): Linear(in_features=65536, out_features=128, bias=True)  
  (2): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (3): ReLU()  
  (4): Dropout(p=0.6, inplace=False)  
  (5): Linear(in_features=128, out_features=128, bias=True)  
  (6): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (7): ReLU()  
  (8): Dropout(p=0.4, inplace=False)  
  (9): Linear(in_features=128, out_features=128, bias=True)  
  (10): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (11): ReLU()  
  (12): Dropout(p=0.3, inplace=False)  
  (13): Linear(in_features=128, out_features=1, bias=True)  
  (14): Sigmoid()  
)
```

```
# hyperparameter  
lr = 0.0001  
weight_decay = 0.0001  
epochs = 10
```

```
Epoch 1/10 - loss: 0.0973 - train_acc: 96.97% - val_loss: 0.3186 - val_acc: 81.25% - time: 62.11s  
Epoch 2/10 - loss: 0.0760 - train_acc: 97.90% - val_loss: 0.3309 - val_acc: 75.00% - time: 63.22s  
Epoch 3/10 - loss: 0.0719 - train_acc: 97.80% - val_loss: 0.1735 - val_acc: 100.00% - time: 61.64s  
Epoch 4/10 - loss: 0.0794 - train_acc: 97.17% - val_loss: 0.1501 - val_acc: 93.75% - time: 63.65s  
Epoch 5/10 - loss: 0.0629 - train_acc: 98.10% - val_loss: 0.1741 - val_acc: 87.50% - time: 62.01s  
Epoch 6/10 - loss: 0.0603 - train_acc: 98.00% - val_loss: 0.1180 - val_acc: 100.00% - time: 63.59s  
Epoch 7/10 - loss: 0.0581 - train_acc: 98.05% - val_loss: 0.3138 - val_acc: 87.50% - time: 63.15s  
Epoch 8/10 - loss: 0.0633 - train_acc: 98.14% - val_loss: 0.2371 - val_acc: 87.50% - time: 62.71s  
Epoch 9/10 - loss: 0.0474 - train_acc: 98.63% - val_loss: 0.6334 - val_acc: 75.00% - time: 63.27s  
Epoch 10/10 - loss: 0.0487 - train_acc: 98.29% - val_loss: 0.1322 - val_acc: 93.75% - time: 61.54s
```



```
Test Accuracy: 83.23%  
Test Loss: 0.4815
```

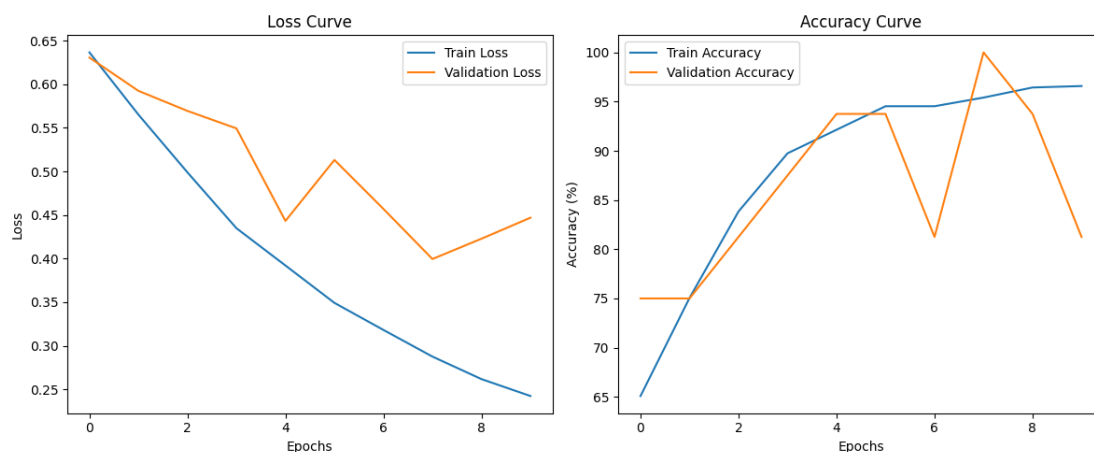
維持前一次(Test 7) model 設定，唯獨改變 weight decay 為 0.0001，然而 validation loss & accuracy 的數值表現並無明顯改善，仍需做其他參數調整改進。

Test 9 :

```
Sequential(  
  (0): Flatten(start_dim=1, end_dim=-1)  
  (1): Linear(in_features=65536, out_features=64, bias=True)  
  (2): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (3): ReLU()  
  (4): Dropout(p=0.6, inplace=False)  
  (5): Linear(in_features=64, out_features=64, bias=True)  
  (6): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (7): ReLU()  
  (8): Dropout(p=0.4, inplace=False)  
  (9): Linear(in_features=64, out_features=64, bias=True)  
  (10): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (11): ReLU()  
  (12): Dropout(p=0.3, inplace=False)  
  (13): Linear(in_features=64, out_features=1, bias=True)  
  (14): Sigmoid()  
)
```

```
# hyperparameter  
lr = 0.0001  
weight_decay = 0.0001  
epochs = 10
```

```
Epoch 1/10 - loss: 0.6366 - train_acc: 65.09% - val_loss: 0.6307 - val_acc: 75.00% - time: 867.22s  
Epoch 2/10 - loss: 0.5652 - train_acc: 75.05% - val_loss: 0.5925 - val_acc: 75.00% - time: 61.85s  
Epoch 3/10 - loss: 0.4990 - train_acc: 83.84% - val_loss: 0.5695 - val_acc: 81.25% - time: 63.32s  
Epoch 4/10 - loss: 0.4349 - train_acc: 89.75% - val_loss: 0.5494 - val_acc: 87.50% - time: 64.03s  
Epoch 5/10 - loss: 0.3921 - train_acc: 92.14% - val_loss: 0.4433 - val_acc: 93.75% - time: 63.27s  
Epoch 6/10 - loss: 0.3492 - train_acc: 94.53% - val_loss: 0.5132 - val_acc: 93.75% - time: 64.99s  
Epoch 7/10 - loss: 0.3181 - train_acc: 94.53% - val_loss: 0.4570 - val_acc: 81.25% - time: 64.09s  
Epoch 8/10 - loss: 0.2876 - train_acc: 95.41% - val_loss: 0.3995 - val_acc: 100.00% - time: 63.00s  
Epoch 9/10 - loss: 0.2617 - train_acc: 96.44% - val_loss: 0.4228 - val_acc: 93.75% - time: 64.23s  
Epoch 10/10 - loss: 0.2423 - train_acc: 96.58% - val_loss: 0.4469 - val_acc: 81.25% - time: 63.08s
```



```
Test Accuracy: 81.61%  
Test Loss: 0.4304
```

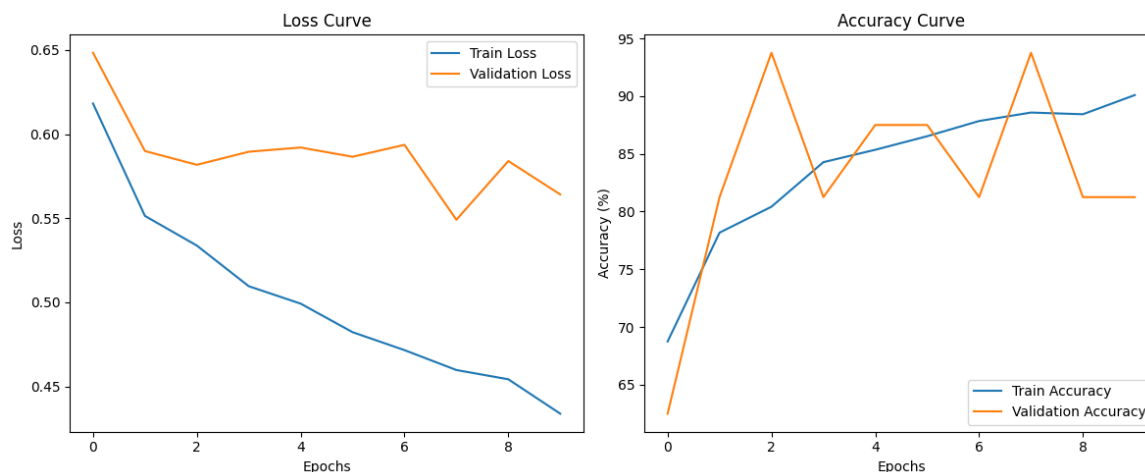
weight decay 與 learning rate 皆維持 0.0001，將每層計算的神經元數從 128 降回 64，training set 表現略降些，但 validation loss 與 validation accuracy 劇烈波動情也有些許改善。目前模型在驗證集上的表現波動還是大，代表模型仍存在過擬合的情況，應再嘗試調整 hyperparameters 設定。

Test 10 :

```
Sequential(  
  (0): Flatten(start_dim=1, end_dim=-1)  
  (1): Linear(in_features=65536, out_features=128, bias=True)  
  (2): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (3): ReLU()  
  (4): Dropout(p=0.6, inplace=False)  
  (5): Linear(in_features=128, out_features=128, bias=True)  
  (6): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (7): ReLU()  
  (8): Dropout(p=0.5, inplace=False)  
  (9): Linear(in_features=128, out_features=128, bias=True)  
  (10): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (11): ReLU()  
  (12): Dropout(p=0.4, inplace=False)  
  (13): Linear(in_features=128, out_features=1, bias=True)  
  (14): Sigmoid()  
)
```

```
# hyperparameter  
lr = 0.00001  
weight_decay = 0.001  
epochs = 10
```

```
Epoch 1/10 - loss: 0.6182 - train_acc: 68.75% - val_loss: 0.6484 - val_acc: 62.50% - time: 519.27s  
Epoch 2/10 - loss: 0.5513 - train_acc: 78.17% - val_loss: 0.5899 - val_acc: 81.25% - time: 60.13s  
Epoch 3/10 - loss: 0.5337 - train_acc: 80.42% - val_loss: 0.5817 - val_acc: 93.75% - time: 61.91s  
Epoch 4/10 - loss: 0.5095 - train_acc: 84.28% - val_loss: 0.5895 - val_acc: 81.25% - time: 60.44s  
Epoch 5/10 - loss: 0.4992 - train_acc: 85.35% - val_loss: 0.5920 - val_acc: 87.50% - time: 63.77s  
Epoch 6/10 - loss: 0.4822 - train_acc: 86.52% - val_loss: 0.5865 - val_acc: 87.50% - time: 60.51s  
Epoch 7/10 - loss: 0.4715 - train_acc: 87.84% - val_loss: 0.5936 - val_acc: 81.25% - time: 60.90s  
Epoch 8/10 - loss: 0.4597 - train_acc: 88.57% - val_loss: 0.5491 - val_acc: 93.75% - time: 61.34s  
Epoch 9/10 - loss: 0.4542 - train_acc: 88.43% - val_loss: 0.5840 - val_acc: 81.25% - time: 62.26s  
Epoch 10/10 - loss: 0.4337 - train_acc: 90.09% - val_loss: 0.5641 - val_acc: 81.25% - time: 62.63s
```



```
Test Accuracy: 83.96%  
Test Loss: 0.5675
```

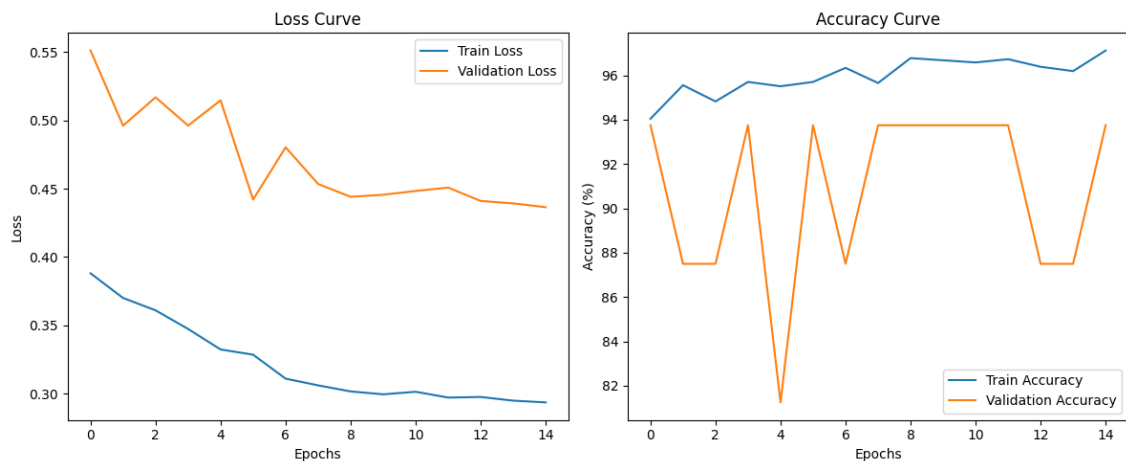
計算 features 全數設為 128 個，dropout 分別依序設為 0.6、0.5、0.4，learning rate 設為 0.00001，weight decay 設為 0.001。

在 validation set 的計算上有些微便穩定成長些，但表現仍欠佳，得到的 test accuracy 為目前為止最佳，可再嘗試改變 dropout 設定，或是嘗試增加訓練 epochs，以得到更佳的 validation set 準確率。

```
Sequential(
  (0): Flatten(start_dim=1, end_dim=-1)
  (1): Linear(in_features=65536, out_features=128, bias=True)
  (2): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (3): ReLU()
  (4): Dropout(p=0.6, inplace=False)
  (5): Linear(in_features=128, out_features=128, bias=True)
  (6): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (7): ReLU()
  (8): Dropout(p=0.4, inplace=False)
  (9): Linear(in_features=128, out_features=128, bias=True)
  (10): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (11): ReLU()
  (12): Dropout(p=0.3, inplace=False)
  (13): Linear(in_features=128, out_features=1, bias=True)
  (14): Sigmoid()
)
```

```
lr = 0.00001
weight_decay = 0.001
epochs = 15
```

```
Epoch 1/15 - loss: 0.3881 - train_acc: 94.04% - val_loss: 0.5512 - val_acc: 93.75% - time: 63.64s
Epoch 2/15 - loss: 0.3700 - train_acc: 95.56% - val_loss: 0.4962 - val_acc: 87.50% - time: 62.60s
Epoch 3/15 - loss: 0.3610 - train_acc: 94.82% - val_loss: 0.5169 - val_acc: 87.50% - time: 64.66s
Epoch 4/15 - loss: 0.3473 - train_acc: 95.70% - val_loss: 0.4963 - val_acc: 93.75% - time: 63.43s
Epoch 5/15 - loss: 0.3323 - train_acc: 95.51% - val_loss: 0.5147 - val_acc: 81.25% - time: 62.12s
Epoch 6/15 - loss: 0.3285 - train_acc: 95.70% - val_loss: 0.4421 - val_acc: 93.75% - time: 63.15s
Epoch 7/15 - loss: 0.3110 - train_acc: 96.34% - val_loss: 0.4803 - val_acc: 87.50% - time: 63.16s
Epoch 8/15 - loss: 0.3061 - train_acc: 95.65% - val_loss: 0.4535 - val_acc: 93.75% - time: 63.23s
Epoch 9/15 - loss: 0.3016 - train_acc: 96.78% - val_loss: 0.4441 - val_acc: 93.75% - time: 62.34s
Epoch 10/15 - loss: 0.2995 - train_acc: 96.68% - val_loss: 0.4456 - val_acc: 93.75% - time: 62.50s
Epoch 11/15 - loss: 0.3014 - train_acc: 96.58% - val_loss: 0.4484 - val_acc: 93.75% - time: 64.67s
Epoch 12/15 - loss: 0.2971 - train_acc: 96.73% - val_loss: 0.4508 - val_acc: 93.75% - time: 61.55s
Epoch 13/15 - loss: 0.2975 - train_acc: 96.39% - val_loss: 0.4411 - val_acc: 87.50% - time: 63.46s
Epoch 14/15 - loss: 0.2949 - train_acc: 96.19% - val_loss: 0.4393 - val_acc: 87.50% - time: 61.81s
Epoch 15/15 - loss: 0.2936 - train_acc: 97.12% - val_loss: 0.4365 - val_acc: 93.75% - time: 64.90s
```



```
Test Accuracy: 83.02%
Test Loss: 0.5005
```

Test 11 :

features 全數設為 128 個，dropout 分別依序設為 0.6、0.4、0.3，learning rate 為 0.00001，weight decay 為 0.001，epochs 數增加至 15，希望得到更穩定計算結果。

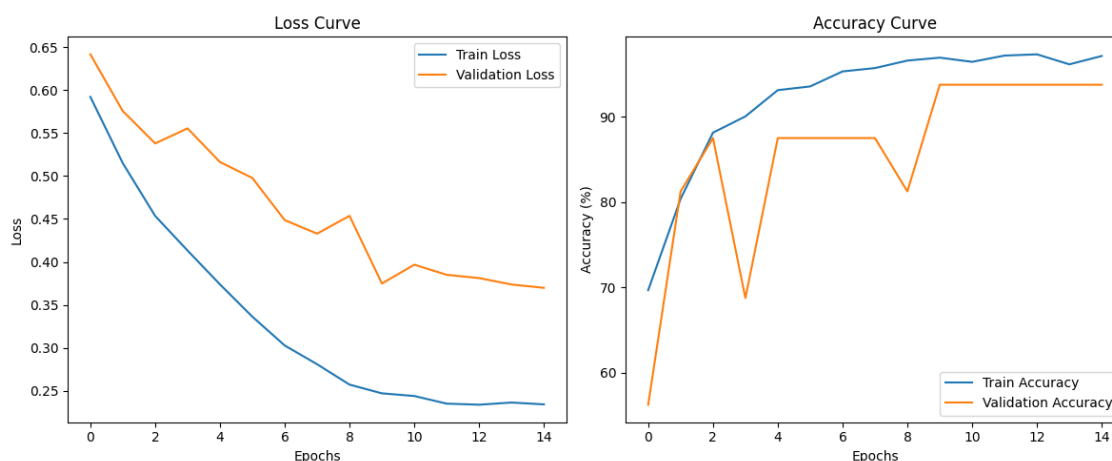
在 training loss 跟 accuracy 確實得到更穩定更佳結果，但在 validation 計算上波動過大，應嘗試減少模型複雜度，或是調整 hyperparameters 優化表現。

Test 12 :

```
Sequential(  
  (0): Flatten(start_dim=1, end_dim=-1)  
  (1): Linear(in_features=65536, out_features=64, bias=True)  
  (2): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (3): ReLU()  
  (4): Dropout(p=0.6, inplace=False)  
  (5): Linear(in_features=64, out_features=128, bias=True)  
  (6): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (7): ReLU()  
  (8): Dropout(p=0.4, inplace=False)  
  (9): Linear(in_features=128, out_features=128, bias=True)  
  (10): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
  (11): ReLU()  
  (12): Dropout(p=0.3, inplace=False)  
  (13): Linear(in_features=128, out_features=1, bias=True)  
  (14): Sigmoid()  
)
```

```
# hyperparameter  
lr = 0.00005  
weight_decay = 0.005  
epochs = 15
```

```
Epoch 1/15 - loss: 0.5923 - train_acc: 69.68% - val_loss: 0.6418 - val_acc: 56.25% - time: 64.25s  
Epoch 2/15 - loss: 0.5151 - train_acc: 80.37% - val_loss: 0.5758 - val_acc: 81.25% - time: 65.56s  
Epoch 3/15 - loss: 0.4536 - train_acc: 88.13% - val_loss: 0.5380 - val_acc: 87.50% - time: 64.86s  
Epoch 4/15 - loss: 0.4133 - train_acc: 90.04% - val_loss: 0.5556 - val_acc: 68.75% - time: 64.86s  
Epoch 5/15 - loss: 0.3739 - train_acc: 93.12% - val_loss: 0.5164 - val_acc: 87.50% - time: 63.44s  
Epoch 6/15 - loss: 0.3362 - train_acc: 93.55% - val_loss: 0.4978 - val_acc: 87.50% - time: 62.62s  
Epoch 7/15 - loss: 0.3027 - train_acc: 95.31% - val_loss: 0.4487 - val_acc: 87.50% - time: 65.37s  
Epoch 8/15 - loss: 0.2808 - train_acc: 95.70% - val_loss: 0.4330 - val_acc: 87.50% - time: 62.98s  
Epoch 9/15 - loss: 0.2571 - train_acc: 96.58% - val_loss: 0.4537 - val_acc: 81.25% - time: 62.90s  
Epoch 10/15 - loss: 0.2469 - train_acc: 96.92% - val_loss: 0.3748 - val_acc: 93.75% - time: 64.14s  
Epoch 11/15 - loss: 0.2438 - train_acc: 96.44% - val_loss: 0.3968 - val_acc: 93.75% - time: 64.10s  
Epoch 12/15 - loss: 0.2349 - train_acc: 97.17% - val_loss: 0.3850 - val_acc: 93.75% - time: 63.70s  
Epoch 13/15 - loss: 0.2337 - train_acc: 97.31% - val_loss: 0.3812 - val_acc: 93.75% - time: 62.22s  
Epoch 14/15 - loss: 0.2362 - train_acc: 96.14% - val_loss: 0.3737 - val_acc: 93.75% - time: 63.37s  
Epoch 15/15 - loss: 0.2341 - train_acc: 97.12% - val_loss: 0.3699 - val_acc: 93.75% - time: 63.97s
```



```
Test Accuracy: 83.02%  
Test Loss: 0.3982
```

features 在不同層運算中有所增加，從 64 增加至 128，dropout 分別依序設為 0.6、0.4、0.3，learning rate 為 0.00005，weight decay 為 0.005，epochs 為 15。

training set 表現進步比且穩定，validation accuracy 相較先前減少一點過大波動情形發生，然而 test accuracy 並未進步，僅降低一些 loss 表現，存在些微過擬合現象，可再嘗試調整 features、learning rate、epochs，或是 batch size 以得到改進 model 表現。

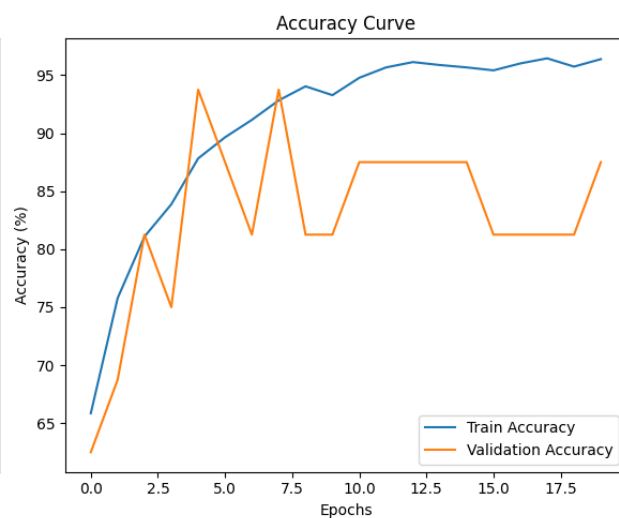
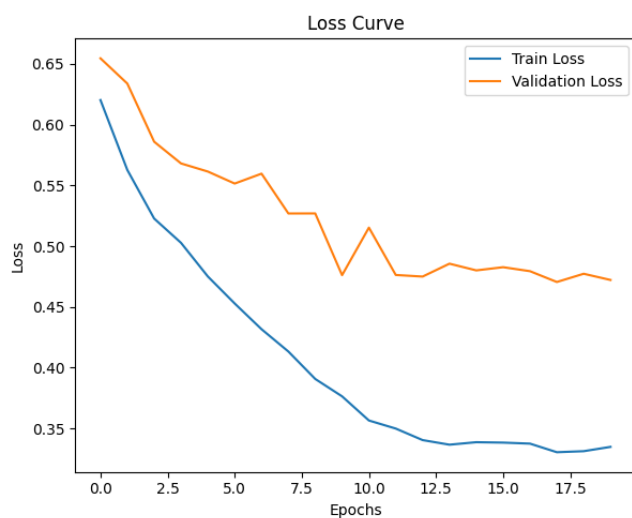
Test 13 :

```
# create dataloaders
train_loader = DataLoader(train_dataset, batch_size=128, shuffle=True)
val_loader = DataLoader(val_dataset, batch_size=128, shuffle=False)
test_loader = DataLoader(test_dataset, batch_size=128, shuffle=False)
```

```
Sequential(
  (0): Flatten(start_dim=1, end_dim=-1)
  (1): Linear(in_features=65536, out_features=64, bias=True)
  (2): BatchNorm1d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (3): ReLU()
  (4): Dropout(p=0.6, inplace=False)
  (5): Linear(in_features=64, out_features=128, bias=True)
  (6): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (7): ReLU()
  (8): Dropout(p=0.4, inplace=False)
  (9): Linear(in_features=128, out_features=128, bias=True)
  (10): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (11): ReLU()
  (12): Dropout(p=0.3, inplace=False)
  (13): Linear(in_features=128, out_features=1, bias=True)
  (14): Sigmoid()
)
```

```
# hyperparameter
lr = 0.00003
weight_decay = 0.005
epochs = 20
```

```
Epoch 1/20 - loss: 0.6202 - train_acc: 65.86% - val_loss: 0.6545 - val_acc: 62.50% - time: 68.02s
Epoch 2/20 - loss: 0.5627 - train_acc: 75.78% - val_loss: 0.6338 - val_acc: 68.75% - time: 63.58s
Epoch 3/20 - loss: 0.5227 - train_acc: 81.08% - val_loss: 0.5859 - val_acc: 81.25% - time: 62.54s
Epoch 4/20 - loss: 0.5028 - train_acc: 83.87% - val_loss: 0.5681 - val_acc: 75.00% - time: 63.29s
Epoch 5/20 - loss: 0.4751 - train_acc: 87.82% - val_loss: 0.5614 - val_acc: 93.75% - time: 61.67s
Epoch 6/20 - loss: 0.4528 - train_acc: 89.64% - val_loss: 0.5515 - val_acc: 87.50% - time: 63.19s
Epoch 7/20 - loss: 0.4317 - train_acc: 91.14% - val_loss: 0.5597 - val_acc: 81.25% - time: 63.32s
Epoch 8/20 - loss: 0.4133 - train_acc: 92.82% - val_loss: 0.5269 - val_acc: 93.75% - time: 62.64s
Epoch 9/20 - loss: 0.3908 - train_acc: 94.03% - val_loss: 0.5269 - val_acc: 81.25% - time: 61.25s
Epoch 10/20 - loss: 0.3765 - train_acc: 93.27% - val_loss: 0.4762 - val_acc: 81.25% - time: 62.73s
Epoch 11/20 - loss: 0.3566 - train_acc: 94.77% - val_loss: 0.5153 - val_acc: 87.50% - time: 63.70s
Epoch 12/20 - loss: 0.3499 - train_acc: 95.66% - val_loss: 0.4763 - val_acc: 87.50% - time: 63.25s
Epoch 13/20 - loss: 0.3405 - train_acc: 96.12% - val_loss: 0.4750 - val_acc: 87.50% - time: 63.07s
Epoch 14/20 - loss: 0.3367 - train_acc: 95.87% - val_loss: 0.4856 - val_acc: 87.50% - time: 63.46s
Epoch 15/20 - loss: 0.3387 - train_acc: 95.66% - val_loss: 0.4801 - val_acc: 87.50% - time: 62.77s
Epoch 16/20 - loss: 0.3384 - train_acc: 95.41% - val_loss: 0.4827 - val_acc: 81.25% - time: 63.36s
Epoch 17/20 - loss: 0.3375 - train_acc: 96.01% - val_loss: 0.4794 - val_acc: 81.25% - time: 63.25s
Epoch 18/20 - loss: 0.3305 - train_acc: 96.45% - val_loss: 0.4705 - val_acc: 81.25% - time: 62.03s
Epoch 19/20 - loss: 0.3313 - train_acc: 95.73% - val_loss: 0.4773 - val_acc: 81.25% - time: 62.85s
Epoch 20/20 - loss: 0.3349 - train_acc: 96.38% - val_loss: 0.4722 - val_acc: 87.50% - time: 64.99s
```



```
Test Accuracy: 80.36%
Test Loss: 0.5629
```

features 設定上維持與前次(**Test 12**)一樣，dropout 依序設為 0.6、0.4、0.3，learning rate 降低為 0.00003，weight decay 為 0.005，epochs 增加為 20，另外將 batch size 調整為 128。然而從結果上觀察 training set 和 validation set 皆未提升表現，計畫藉由調整 features、epochs 數量改善降低 loss、改善 validation accuracy 波動明顯的問題。

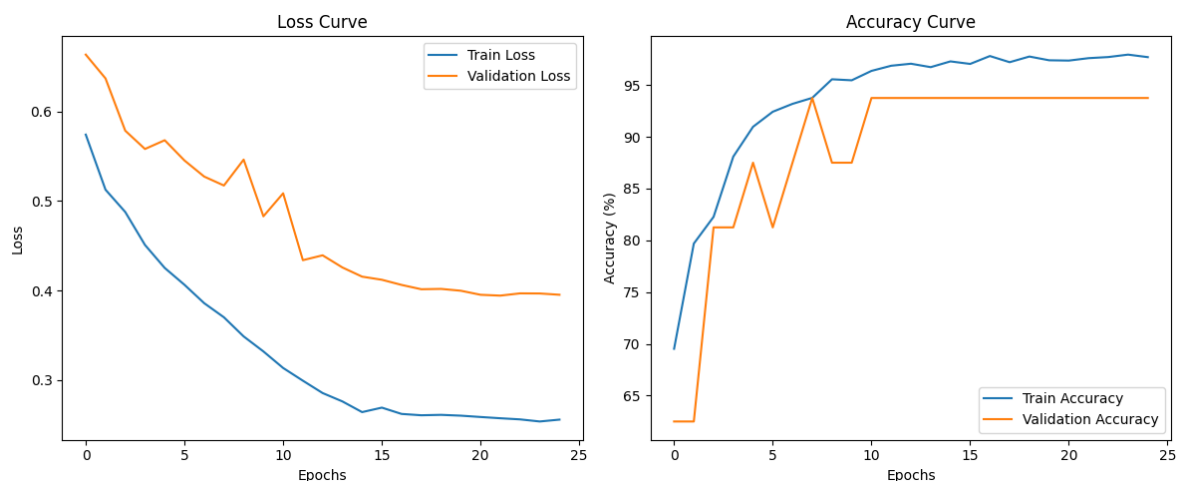
Test 14：

```
# create dataloaders
train_loader = DataLoader(train_dataset, batch_size=128, shuffle=True)
val_loader = DataLoader(val_dataset, batch_size=128, shuffle=False)
test_loader = DataLoader(test_dataset, batch_size=128, shuffle=False)
```

```
Sequential(
  (0): Flatten(start_dim=1, end_dim=-1)
  (1): Linear(in_features=65536, out_features=128, bias=True)
  (2): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (3): ReLU()
  (4): Dropout(p=0.6, inplace=False)
  (5): Linear(in_features=128, out_features=128, bias=True)
  (6): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (7): ReLU()
  (8): Dropout(p=0.4, inplace=False)
  (9): Linear(in_features=128, out_features=128, bias=True)
  (10): BatchNorm1d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (11): ReLU()
  (12): Dropout(p=0.3, inplace=False)
  (13): Linear(in_features=128, out_features=1, bias=True)
  (14): Sigmoid()
)
```

```
# hyperparameter
lr = 0.00003
weight_decay = 0.005
epochs = 25
```

```
Epoch 1/25 - loss: 0.5741 - train_acc: 69.52% - val_loss: 0.6635 - val_acc: 62.50% - time: 67.33s
Epoch 2/25 - loss: 0.5126 - train_acc: 79.69% - val_loss: 0.6369 - val_acc: 62.50% - time: 62.90s
Epoch 3/25 - loss: 0.4877 - train_acc: 82.27% - val_loss: 0.5785 - val_acc: 81.25% - time: 62.11s
Epoch 4/25 - loss: 0.4509 - train_acc: 88.09% - val_loss: 0.5581 - val_acc: 81.25% - time: 62.09s
Epoch 5/25 - loss: 0.4252 - train_acc: 90.97% - val_loss: 0.5679 - val_acc: 87.50% - time: 60.51s
Epoch 6/25 - loss: 0.4064 - train_acc: 92.42% - val_loss: 0.5453 - val_acc: 81.25% - time: 62.10s
Epoch 7/25 - loss: 0.3858 - train_acc: 93.18% - val_loss: 0.5273 - val_acc: 87.50% - time: 60.50s
Epoch 8/25 - loss: 0.3700 - train_acc: 93.76% - val_loss: 0.5173 - val_acc: 93.75% - time: 63.75s
Epoch 9/25 - loss: 0.3486 - train_acc: 95.56% - val_loss: 0.5462 - val_acc: 87.50% - time: 60.67s
Epoch 10/25 - loss: 0.3319 - train_acc: 95.46% - val_loss: 0.4829 - val_acc: 87.50% - time: 61.79s
Epoch 11/25 - loss: 0.3134 - train_acc: 96.37% - val_loss: 0.5086 - val_acc: 93.75% - time: 60.62s
Epoch 12/25 - loss: 0.2992 - train_acc: 96.87% - val_loss: 0.4338 - val_acc: 93.75% - time: 63.39s
Epoch 13/25 - loss: 0.2854 - train_acc: 97.06% - val_loss: 0.4393 - val_acc: 93.75% - time: 60.53s
Epoch 14/25 - loss: 0.2760 - train_acc: 96.73% - val_loss: 0.4257 - val_acc: 93.75% - time: 61.20s
Epoch 15/25 - loss: 0.2641 - train_acc: 97.29% - val_loss: 0.4154 - val_acc: 93.75% - time: 61.05s
Epoch 16/25 - loss: 0.2691 - train_acc: 97.04% - val_loss: 0.4120 - val_acc: 93.75% - time: 60.39s
Epoch 17/25 - loss: 0.2620 - train_acc: 97.80% - val_loss: 0.4062 - val_acc: 93.75% - time: 63.23s
Epoch 18/25 - loss: 0.2605 - train_acc: 97.21% - val_loss: 0.4014 - val_acc: 93.75% - time: 60.48s
Epoch 19/25 - loss: 0.2610 - train_acc: 97.75% - val_loss: 0.4017 - val_acc: 93.75% - time: 61.82s
Epoch 20/25 - loss: 0.2601 - train_acc: 97.39% - val_loss: 0.3997 - val_acc: 93.75% - time: 60.37s
Epoch 21/25 - loss: 0.2586 - train_acc: 97.36% - val_loss: 0.3952 - val_acc: 93.75% - time: 61.17s
Epoch 22/25 - loss: 0.2572 - train_acc: 97.60% - val_loss: 0.3942 - val_acc: 93.75% - time: 62.74s
Epoch 23/25 - loss: 0.2559 - train_acc: 97.71% - val_loss: 0.3967 - val_acc: 93.75% - time: 61.45s
Epoch 24/25 - loss: 0.2536 - train_acc: 97.94% - val_loss: 0.3966 - val_acc: 93.75% - time: 61.22s
Epoch 25/25 - loss: 0.2557 - train_acc: 97.70% - val_loss: 0.3952 - val_acc: 93.75% - time: 60.66s
```



Test Accuracy: 81.32%
Test Loss: 0.4571

Features 每層皆設為 128，dropout 依序設為 0.6、0.4、0.3，learning rate 為 0.00003，weight decay 為 0.005，epochs 增加為 25，batch size 為 128。

增加 epochs 數量增加了後期運算上有效降低 loss、穩定的 accuracy，然而 train loss 跟 validation loss 還有再降低的改善空間。

推測還可以再改進的部分包括將訓練 epochs 增加，然而持續增加 epochs 將會使訓練 model 所花時間大大增加；或是調整增加 hidden layer 個數，增加 model 深度；抑或是嘗試改變使用的 activation function；也可考慮進行 data augmentation，提高 model 的 generalization ability。